Assignment

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2024-01-31

#question 1

```
p <- 0.5
N_values <- c(10, 50, 500)

calculate_ks_distance <- function(N) {

  Y <- rbinom(1000, N, p)


  cdf_Y <- ecdf(Y)


  clt_samples <- replicate(1000, sum(rbinom(N, 1, p)) - N*p)

  cdf_CLT <- ecdf(clt_samples)

  ks_distance <- ks.test(Y, clt_samples)$statistic

  return(ks_distance)
}

ks_distances <- sapply(N_values, calculate_ks_distance)
```

```
## Warning in ks.test.default(Y, clt_samples): p-value will be approximate in the
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```
for (i in seq_along(N_values)) {
  cat("KS Distance for N =", N_values[i], ":", ks_distances[i], "\n")
}
```

```
## KS Distance for N = 10 : 0.886
## KS Distance for N = 50 : 1
## KS Distance for N = 500 : 1
```

#question 3

```
integration<-function(x)
{
```

```r
  root_x = 2*sqrt(x)
  bessel<-2*besselK(root_x, nu = 0)
  return(bessel)
}

integrate(integration,0,Inf)
```

```
## 1 with absolute error < 8.1e-06
```